US Real Estate CRM

Complete Product & Technical Documentation

Field	Details
Project Type	SaaS-ready Real Estate Customer Relationship Management System
Target Market	United States Real Estate Agencies, Brokers and Agents
Initial Version	MVP
Primary Goal	Build a professional, scalable CRM that manages real estate leads, clients, properties, activities, deals, and agent performance from one centralized platform.

Table of Contents

Table of Contents 2
1. Project Overview 6
2. Business Objectives 6
Primary Objectives 6
Long-Term Business Goal 6
3. Target Users 7
3.1 Super Admin 7
3.2 Agency Admin 7
3.3 Manager / Broker 7
3.4 Real Estate Agent 7
4. Core Modules 8
Future Modules 8
5. Authentication & Authorization 8
Features 8
Roles 9
6. Dashboard 9
KPI Cards 9
Charts 9
Recent Activity 10
7. Lead Management 10
Lead Information 10
Lead Sources 10
Lead Status 11
Lead Operations 11
8. Client Management 11
Client Types 11
Client Information 12
Client History 12
9. Property Management 12
Property Information 12
Property Types 13
Property Status 13
Property Features 13

10. Sales Pipeline 13
11. Deal Management 14
Deal Information 14
Deal Status 14
12. Tasks & Follow-ups 15
Task Types 15
Task Information 15
Task Status 15
13. Appointments 15
Appointment Information 15
Future Version 16
14. Notes & Activity Timeline 16
15. Notifications 16
Notification Channels 17
Future 17
16. Agent Management 17
Agent Profile 17
Performance Metrics 17
17. Reports & Analytics 18
Lead Reports 18
Property Reports 18
Sales Reports 18
Agent Reports 18
Future 18
18. Search & Filtering 18
19. Database Design 19
20. Basic Database Relationships 20
Lead 20
Client 20
Deal 20
21. Technology Stack 20
Backend 20
Frontend 21
Authentication 21
Background Processing 21

File Storage 21
Deployment 21
CI/CD 21
22. Recommended Project Architecture 21
Backend 21
23. API Structure 22
Authentication 22
Leads 22
Clients 22
Properties 23
Deals 23
24. Security Requirements 23
25. SaaS Architecture — Future 23
26. Subscription Model — Future 24
27. AI Roadmap 24
AI Lead Scoring 24
AI Lead Summary 25
AI Follow-up Generator 25
Property Recommendation 25
AI Chatbot 25
28. MVP Scope 25
MVP Must Have 25
Phase 2 26
Phase 3 26
29. Development Roadmap 26
Sprint 1 — Project Setup 26
Sprint 2 — Authentication 26
Sprint 3 — Lead Management 27
Sprint 4 — Client Management 27
Sprint 5 — Property Management 27
Sprint 6 — Sales Pipeline 27
Sprint 7 — Tasks & Appointments 27
Sprint 8 — Dashboard & Reports 28
Sprint 9 — Notifications 28
Sprint 10 — Testing 28

Sprint 11 — Deployment 28
Sprint 12 — MVP Launch 28
30. Testing Strategy 29
Unit Testing 29
API Testing 29
Integration Testing 29
Security Testing 29
User Acceptance Testing 29
31. DevOps Strategy 30
32. Documentation Required 30
33. Success Metrics 31
34. Future Expansion 31
35. Final Product Vision 32
36. Immediate Next Step 32
Step 1 32
Step 2 32
Step 3 32
Step 4 32
Step 5 32
Step 6 32
Step 7 33
Step 8 33

1. Project Overview

The US Real Estate CRM will be a web-based platform designed to help real estate agencies and agents manage their complete customer and sales lifecycle.

The system will centralize:

- Leads
- Clients
- Properties
- Agents
- Follow-ups
- Tasks
- Appointments
- Sales pipeline
- Deals
- Communication history
- Reports and analytics

The first objective is to develop a professional MVP (Minimum Viable Product) that can be used for real-world testing.

The architecture will be designed from the beginning with scalability in mind so the platform can later become a multi-tenant SaaS product.

2. Business Objectives

Primary Objectives

- Centralize real estate business operations.
- Improve lead management.
- Prevent missed follow-ups.
- Improve agent productivity.
- Track the complete sales pipeline.
- Monitor property and client activity.
- Provide management-level reporting.
- Reduce manual administrative work.
- Introduce automation.
- Create a foundation for a future SaaS business.

Long-Term Business Goal

Convert the CRM into a subscription-based SaaS platform where multiple real estate companies can create their own accounts and independently manage their agents, clients, properties, and leads.

3. Target Users

3.1 Super Admin

Platform owner.

Responsibilities:

- Manage organizations
- Manage subscriptions
- Manage platform users
- Monitor system activity
- Manage plans
- View platform analytics
- Manage system settings

3.2 Agency Admin

Owner/administrator of a real estate company.

Responsibilities:

- Manage agents
- Manage leads
- Manage clients
- Manage properties
- Assign leads
- Monitor sales
- View reports
- Manage company settings

3.3 Manager / Broker

Responsibilities:

- Monitor agents
- Assign leads
- Track deals
- Review performance
- View reports
- Manage property inventory

3.4 Real Estate Agent

Responsibilities:

- Manage assigned leads
- Manage clients
- Add properties
- Schedule appointments
- Add notes
- Track follow-ups
- Update deal stages

4. Core Modules

The MVP will contain the following major modules:

- Authentication & Authorization
- Dashboard
- Lead Management
- Client Management
- Property Management
- Agent Management
- Sales Pipeline
- Deal Management
- Tasks & Follow-ups
- Appointments
- Notes & Activity History
- Notifications
- Reports & Analytics
- Settings

Future Modules

- Email Integration
- SMS Integration
- WhatsApp Integration
- AI Features
- Subscription & Billing
- Multi-Tenant SaaS Management

5. Authentication & Authorization

Features

- User registration

- Login
- Logout
- Forgot password
- Reset password
- Email verification
- Profile management
- Change password
- Role-based access control
- JWT authentication for APIs

Roles

```
Super Admin
    |
Agency Admin
    |
Manager / Broker
    |
Agent
```

Each role will have different permissions.

6. Dashboard

The dashboard will provide an overview of business activity.

KPI Cards

- Total Leads
- New Leads
- Active Clients
- Total Properties
- Active Properties
- Open Deals
- Closed Deals
- Total Revenue

Charts

- Leads by month
- Deals by month
- Revenue by month
- Lead sources
- Deal pipeline

- Agent performance

Recent Activity

Display:

- New leads
- Updated leads
- New clients
- New properties
- Upcoming appointments
- Completed deals

7. Lead Management

Lead management will be one of the most important CRM modules.

Lead Information

Each lead may contain:

- First Name
- Last Name
- Email
- Phone
- Lead Source
- Lead Type
- Assigned Agent
- Status
- Priority
- Budget
- Property Preference
- Location Preference
- Notes
- Created Date
- Last Contact Date
- Next Follow-up Date

Lead Sources

Examples:

- Website
- Zillow

- Realtor.com
- Facebook
- Instagram
- Google Ads
- Referral
- Phone
- Email
- Walk-in
- Other

Lead Status

```
New
Contacted
Qualified
Unqualified
Appointment Scheduled
Property Viewing
Negotiation
Converted
Lost
```

Lead Operations

- Create lead
- Update lead
- Delete/archive lead
- Assign agent
- Change status
- Add notes
- Schedule follow-up
- Convert lead to client
- View activity history

8. Client Management

A lead can become a client after qualification.

Client Types

- Buyer
- Seller
- Investor

- Renter
- Landlord

Client Information

- Name
- Email
- Phone
- Address
- Client type
- Budget
- Preferred location
- Property type
- Bedrooms
- Bathrooms
- Notes
- Assigned agent

Client History

The system should show:

- Leads
- Properties viewed
- Appointments
- Calls
- Emails
- Notes
- Deals
- Follow-ups

9. Property Management

Agents and agencies can manage their property inventory.

Property Information

- Property title
- Property type
- Address
- City
- State
- ZIP Code

- Price
- Bedrooms
- Bathrooms
- Square footage
- Lot size
- Year built
- Description
- Images
- Agent
- Status

Property Types

- Single Family
- Condo
- Townhouse
- Multi-Family
- Land
- Commercial
- Apartment

Property Status

```
Draft
Active
Pending
Under Contract
Sold
Off Market
Rented
```

Property Features

- Image gallery
- Property details
- Agent assignment
- Client matching
- Activity history

10. Sales Pipeline

The CRM will provide a visual pipeline for tracking deals.

Example:

```
New Lead
    ↓
Contacted
    ↓
Qualified
    ↓
Property Viewing
    ↓
Offer
    ↓
Negotiation
    ↓
Under Contract
    ↓
Closed
```

Agents will be able to move deals between stages.

Managers will be able to see the overall pipeline.

11. Deal Management

A deal connects:

```
Client + Property + Agent + Transaction
```

Deal Information

- Deal name
- Client
- Property
- Agent
- Deal stage
- Estimated value
- Actual value
- Expected closing date
- Closing date
- Commission
- Notes
- Status

Deal Status

- Open
- Won
- Lost

- Cancelled

12. Tasks & Follow-ups

The CRM will help agents avoid missing important activities.

Task Types

- Call
- Email
- Follow-up
- Meeting
- Property Viewing
- Document Review
- Other

Task Information

- Task title
- Description
- Assigned user
- Related lead/client
- Due date
- Priority
- Status

Task Status

```
Pending
In Progress
Completed
Cancelled
```

13. Appointments

Agents can schedule:

- Client meetings
- Property viewings
- Calls
- Consultations
- Follow-ups

Appointment Information

- Title
- Client
- Agent
- Property
- Date
- Start time
- End time
- Location
- Notes
- Status

Future Version

- Google Calendar integration
- Outlook Calendar integration

14. Notes & Activity Timeline

Every lead/client should have an activity timeline.

Example:

```
25 Aug
Lead created

25 Aug
Assigned to John

26 Aug
Phone call completed

27 Aug
Property viewing scheduled

29 Aug
Client viewed property

30 Aug
Offer submitted
```

This provides complete customer history.

15. Notifications

The system should notify users about important activities.

Examples:

- New lead assigned
- Follow-up due
- Appointment approaching
- Deal status changed
- Task overdue
- New client assigned

Notification Channels

- In-app notifications
- Email notifications

Future

- SMS
- WhatsApp

16. Agent Management

Managers can manage their team.

Agent Profile

- Name
- Email
- Phone
- Profile picture
- Role
- Assigned leads
- Active clients
- Properties
- Deals
- Revenue
- Performance

Performance Metrics

- Leads handled
- Leads converted
- Deals closed
- Revenue generated
- Conversion rate
- Follow-ups completed

17. Reports & Analytics

The reporting module will provide management insights.

Lead Reports

- Total leads
- Leads by source
- Leads by status
- Leads by agent
- Conversion rate

Property Reports

- Active properties
- Sold properties
- Pending properties
- Properties by type
- Properties by location

Sales Reports

- Open deals
- Closed deals
- Lost deals
- Revenue
- Commission

Agent Reports

- Leads assigned
- Leads converted
- Deals closed
- Revenue
- Performance comparison

Future

- Export PDF
- Export Excel/CSV
- Scheduled reports

18. Search & Filtering

The CRM should provide global search.

Users should be able to search:

- Leads
- Clients
- Properties
- Agents
- Deals

Filters should include:

- Status
- Agent
- Date
- Location
- Property type
- Price
- Lead source

19. Database Design

Initial database entities:

```
User
Organization
Role
Lead
Client
Property
PropertyImage
Deal
PipelineStage
Task
Appointment
Note
Activity
Notification
AgentProfile
```

Future SaaS entities:

```
Subscription
SubscriptionPlan
Invoice
Payment
Tenant
Usage
```

20. Basic Database Relationships

```
Organization
    |
    +---- Users
    |
    +---- Leads
    |
    +---- Clients
    |
    +---- Properties
    |
    +---- Deals
    |
    +---- Tasks
    |
    +---- Appointments
```

Lead

```
Lead
 ├── assigned_agent
 ├── activities
 ├── notes
 ├── tasks
 └── client
```

Client

```
Client
 ├── properties
 ├── appointments
 ├── deals
 ├── notes
 └── activities
```

Deal

```
Deal
 ├── client
 ├── property
 ├── agent
 └── pipeline_stage
```

21. Technology Stack

Backend

- Python
- Django
- Django REST Framework
- PostgreSQL

Frontend

- React.js
- Tailwind CSS

Authentication

- JWT
- Role-Based Access Control

Background Processing

- Celery
- Redis

File Storage

Initial: Local / Cloud Storage

Production: Azure Blob Storage

Deployment

- Docker
- Microsoft Azure

CI/CD

- Git
- GitHub
- GitHub Actions

22. Recommended Project Architecture

Backend

```
realestate-crm/
│
├── backend/
│   ├── config/
│   ├── accounts/
│   ├── organizations/
```

```
│   ├── leads/
│   ├── clients/
│   ├── properties/
│   ├── deals/
│   ├── tasks/
│   ├── appointments/
│   ├── notifications/
│   ├── reports/
│   └── common/
│
├── frontend/
│   ├── src/
│   │   ├── components/
│   │   ├── pages/
│   │   ├── layouts/
│   │   ├── services/
│   │   ├── hooks/
│   │   └── utils/
│
├── docker/
├── docs/
├── .github/
├── docker-compose.yml
└── README.md
```

23. API Structure

Base URL:

```
/api/v1/
```

Authentication

```
POST /api/v1/auth/login/
POST /api/v1/auth/register/
POST /api/v1/auth/logout/
POST /api/v1/auth/refresh/
POST /api/v1/auth/forgot-password/
```

Leads

```
GET    /api/v1/leads/
POST   /api/v1/leads/
GET    /api/v1/leads/{id}/
PUT    /api/v1/leads/{id}/
DELETE /api/v1/leads/{id}/
```

Clients

```
GET    /api/v1/clients/
```

```
POST   /api/v1/clients/
GET    /api/v1/clients/{id}/
PUT    /api/v1/clients/{id}/
DELETE /api/v1/clients/{id}/
```

Properties

```
GET    /api/v1/properties/
POST   /api/v1/properties/
GET    /api/v1/properties/{id}/
PUT    /api/v1/properties/{id}/
DELETE /api/v1/properties/{id}/
```

Deals

```
GET    /api/v1/deals/
POST   /api/v1/deals/
GET    /api/v1/deals/{id}/
PUT    /api/v1/deals/{id}/
```

24. Security Requirements

The application must implement:

- Secure authentication
- Password hashing
- JWT expiration
- Role-based permissions
- Object-level permissions
- HTTPS in production
- CSRF protection where applicable
- CORS configuration
- Input validation
- API throttling/rate limiting
- Secure file uploads
- Environment variables for secrets
- Database backup strategy
- Audit logging

Sensitive configuration should never be committed to GitHub.

25. SaaS Architecture — Future

The system should eventually support multiple organizations.

Example:

```
CRM Platform
│
├── Company A
│    ├── Admin
│    ├── Agents
│    ├── Leads
│    └── Properties
│
├── Company B
│    ├── Admin
│    ├── Agents
│    ├── Leads
│    └── Properties
│
└── Company C
     ├── Admin
     ├── Agents
     ├── Leads
     └── Properties
```

Each organization must only access its own data.

This is called multi-tenancy.

26. Subscription Model — Future

Potential plans:

Plan	Price	Includes
Free	$0	1 User, 50 Leads, 10 Properties, Basic Dashboard
Starter	$29/month	5 Users, 500 Leads, 100 Properties, Reports, Email Notifications
Professional	$79/month	15 Users, Unlimited Leads, Unlimited Properties, Advanced Reports, Automation, AI Features
Agency	$149+/month	Unlimited Users, Advanced Analytics, Automation, AI, Priority Support, Custom Features

These prices are initial business assumptions and should be validated before launch.

27. AI Roadmap

AI should not be the first development priority.

First build a stable CRM. Then introduce:

AI Lead Scoring

Predict which leads are more likely to convert.

AI Lead Summary

Generate a short summary from the customer's history.

AI Follow-up Generator

Generate professional follow-up messages.

Property Recommendation

Recommend properties based on:

- Budget
- Location
- Property type
- Bedrooms
- Bathrooms
- Client preferences

AI Chatbot

Allow website visitors to ask:

```
"Show me 3-bedroom houses in Texas under $400,000."
```

The chatbot can search available CRM properties.

28. MVP Scope

The first version should NOT include everything.

MVP Must Have

- Authentication
- User roles
- Dashboard
- Lead management
- Client management
- Property management
- Agent management
- Sales pipeline
- Deal management
- Tasks

- Follow-ups
- Appointments
- Notes
- Notifications
- Basic reports
- Search/filtering

Phase 2

- Email integration
- Calendar integration
- Advanced reports
- Automation
- Import/export
- Cloud storage

Phase 3

- AI features
- SMS
- WhatsApp
- AI chatbot
- Subscription billing
- Multi-tenant SaaS
- Mobile application

29. Development Roadmap

Sprint 1 — Project Setup

- Git repository
- Django setup
- React setup
- PostgreSQL
- Environment configuration
- Docker
- Basic project architecture

Sprint 2 — Authentication

- User model
- Login

- Registration
- JWT
- Roles
- Permissions
- Profile

Sprint 3 — Lead Management

- Lead model
- Lead API
- Lead dashboard
- Lead assignment
- Lead status
- Lead activity

Sprint 4 — Client Management

- Client model
- Client API
- Client profile
- Lead-to-client conversion
- Client activity timeline

Sprint 5 — Property Management

- Property model
- Property CRUD
- Property images
- Property search
- Property filtering
- Agent assignment

Sprint 6 — Sales Pipeline

- Pipeline stages
- Deals
- Deal management
- Kanban board
- Deal tracking

Sprint 7 — Tasks & Appointments

- Tasks

- Follow-ups
- Appointments
- Calendar UI
- Reminders

Sprint 8 — Dashboard & Reports

- KPIs
- Charts
- Agent reports
- Lead reports
- Sales reports

Sprint 9 — Notifications

- In-app notifications
- Email notifications
- Celery
- Redis

Sprint 10 — Testing

- Unit tests
- API tests
- Integration tests
- Permission tests
- Frontend testing
- Security testing

Sprint 11 — Deployment

- Docker production configuration
- Azure deployment
- PostgreSQL production database
- Blob storage
- HTTPS
- Environment variables
- CI/CD

Sprint 12 — MVP Launch

- Production testing
- Bug fixing

- Documentation
- Demo account
- User feedback
- Performance optimization

30. Testing Strategy

Testing will be performed at multiple levels.

Unit Testing

Test individual functions and business logic.

API Testing

Test:

- Authentication
- CRUD operations
- Permissions
- Validation
- Error handling

Integration Testing

Test interactions between:

```
Lead → Client → Property → Deal
```

Security Testing

Test:

- Unauthorized access
- Role permissions
- API authentication
- Data isolation
- File upload security

User Acceptance Testing

Real users should test:

- Lead creation
- Lead assignment
- Property management

- Client management
- Deal management
- Reports

31. DevOps Strategy

Development workflow:

```
Developer
    ↓
Git
    ↓
GitHub
    ↓
Pull Request
    ↓
Automated Tests
    ↓
Build Docker Image
    ↓
CI/CD
    ↓
Azure
    ↓
Production
```

Environments:

```
Development
     ↓
Staging
     ↓
Production
```

32. Documentation Required

The project should maintain:

- README.md
- API Documentation
- Database Documentation
- Installation Guide
- Deployment Guide
- Environment Configuration
- User Guide
- Admin Guide
- Developer Guide
- Testing Documentation

- Architecture Documentation

API documentation can later be provided using: Swagger / OpenAPI

33. Success Metrics

The project should eventually measure:

- Number of active users
- Number of leads managed
- Lead conversion rate
- Deals closed
- Revenue generated
- Average response time
- Follow-up completion rate
- Monthly active organizations
- Customer retention
- Monthly recurring revenue

34. Future Expansion

After establishing the CRM, the platform can expand into:

```
Real Estate CRM
       ↓
Real Estate SaaS
       ↓
AI Real Estate Platform
       ↓
Lead Generation
       ↓
Marketing Automation
       ↓
Property Management
```

Potential future integrations:

- Real estate listing platforms
- Google Maps
- Google Calendar
- Microsoft Outlook
- Email providers
- SMS providers
- Payment gateways
- AI APIs

35. Final Product Vision

The final product should become more than a simple CRUD CRM.

The long-term vision is:

> ***A complete AI-powered Real Estate CRM for US real estate professionals that helps agencies capture leads, manage clients, manage properties, automate follow-ups, track deals, analyze performance, and increase sales.***

The development strategy will be:

```
MVP
 ↓
Real-world testing
 ↓
User feedback
 ↓
Automation
 ↓
AI
 ↓
Multi-tenancy
 ↓
Subscription billing
 ↓
SaaS product
```

36. Immediate Next Step

Development should begin with:

Step 1

Finalize requirements.

Step 2

Design database/ERD.

Step 3

Create GitHub repository.

Step 4

Set up Django + DRF + PostgreSQL.

Step 5

Set up React + Tailwind.

Step 6

Set up Docker.

Step 7

Implement authentication and role permissions.

Step 8

Start the Lead Management module.

> **Important:** We should build the MVP first rather than starting with AI, payments, WhatsApp, or other advanced features. This keeps the project manageable and allows us to validate the core CRM before investing in advanced functionality.